

Lab Handout # 4

Li-Fi(Light Fidelity) data transceiver using On chip ROM(Read only memory)

Design and Project LAB

Dept. of ECE, The LNMIIT, Jaipur

Time : 3:00 Hour

Maximum Marks : 10

Instructions and information for students

- This Lab Handout consists of 2 pages. Please check that you have a complete copy.
- Simulate in Vivado Software.

Objective:

- 1) Li-Fi Transceiver.

Simulation and implementation of Li-Fi Transceiver using Evaluation board

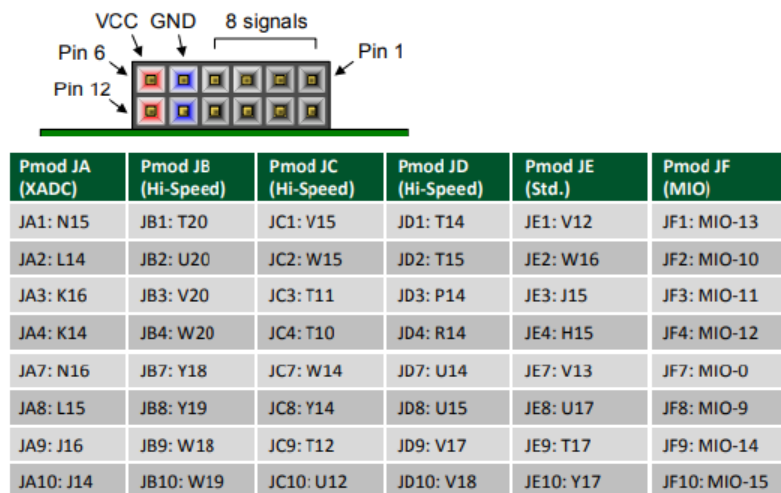


Fig. Pmod Pin Diagram

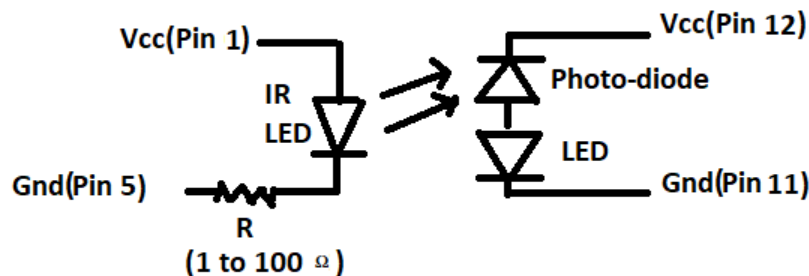


Fig. 1. Li-Fi IR TRX.

1) Introduction

- A Li-Fi transmitter uses visible/invisible light for data transmission using on-off keying (OOK).
- The on chip memory of ZYBO board is utilized in Read only mode to store data (in a coefficient file "*.coe") with various configurations.
- An IR-Photodiode pair is utilized for data transmission (which is stored in the ROM).

- The General purpose Input/output (GPIO) pins are utilized for data transfer between evaluation board and the IR ckt.

2) Procedure: Simulation & Implementation

- Use "Block Memory Generator" IP(256 KB Internal on chip RAM/ROM) to include ROM(Read Only Memory) in the hardware design.
- PORT map the above IP using Component Instantiation.
- Configure the memory for various storage configurations like 16 bit width(WORD) with 16 depth(no. of storage elements), 8 bit (BYTE) width with 32 depth, 4 bit width with 64 depth etc..
- Use clock to drive the memory read operation.
- Use the on board GPIO pins (Pmod JC) for data transfer between the evaluation board and ckt. as shown in the figure.
- Connect the IR tx. and Rx. ckt as shown in the figure on the breadboard.(The longer leg is anode i.e. +ve).

3) Observations and Results.

- Implement the above design on ZYBO board with memory read operation. Use GPIO and IR TRX to blink LED on the receiver. [2]
- Simulate and Implement ROM with 16 bit word storage of 16 array size. [2]
- Use GPIO and IR TRX to blink LED on the receiver, with memory read speeds of 1 second, 0.5 second, 0.25 second durations. [2]
- Blink the receiver LED for the LSB of the data stored in the ROM. [4]

WELL DONE
